

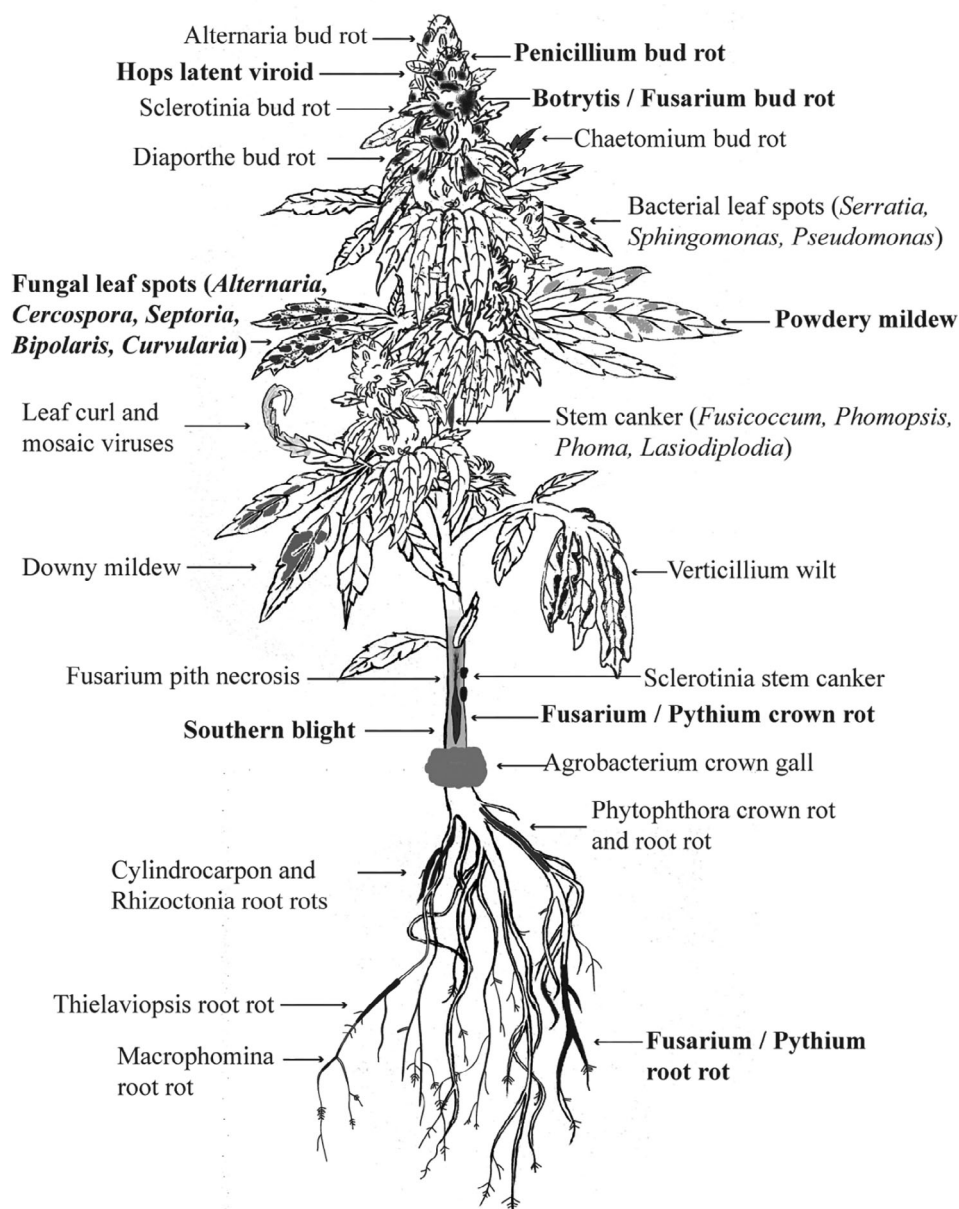


Figure 2. The emerging pathogens on cannabis and hemp plants. Pathogens shown in **bold** are the most damaging.

polymerase chain reaction (PCR) of the ribosomal DNA region that includes the internal transcribed spacer (ITS) and intergenic spacer regions (IGS)^{34–37} (Fig. S2). PCR was used to identify most of the species listed in Table S1. In addition, the elongation factor 1 (EF-1) region was used to discriminate among *Fusarium* species.^{21,38} For *Golovinomyces* causing powdery mildew and *Botrytis* species causing bud rot, additional molecular markers are required to differentiate between species. In the latter, the glyceraldehyde-3-phosphate dehydrogenase gene (G3PDH) and the heat shock protein 60 gene (HSP60) were used.^{39,40} For the powdery mildew pathogen *Golovinomyces*, however, the ITS region was insufficient to distinguish among species.⁴¹ Additional molecular markers are needed to conclusively confirm or amalgamate the three species currently reported to cause powdery mildew on cannabis and hemp (Table S1). Following identification, demonstration of pathogenicity on cannabis and hemp

plants is an essential requirement – reports of presence or recovery of fungi from these tissues is insufficient, since many saprophytes are found on cannabis and hemp plants. Therefore, fulfillment of Koch's postulates²⁶ is essential. Deposition of DNA sequences in a repository such as GenBank will also allow researchers to compare their findings with others. Subspecies or *formae specialis* designations, e.g. *f.sp. cannabis* or *cannabina* for pathogens recovered from cannabis or hemp plants, is discouraged until host range studies and molecular confirmations support these designations. The host of origin is insufficient to imply specific host preference since most of the pathogens affecting cannabis and hemp are known to have wide host ranges (Table S1).

Bacterial plant pathogens have also been identified based on PCR methods that utilize the 16S region of ribosomal RNA (rRNA) in addition to other methods.^{35,42,43} To date, recent reports of